Machine Learning

Matrix Factorization

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Data
Intelligence
LAB

Problem Setting

- Suppose we have a **sparse** user-item rating matrix
- Which rating will be given on each empty cell?

	5		8	7		
	9	10	8	9		
	6	6		6	10	7
	8				6	10

Problem Setting

- **This is a typical recommendation scenario**
- **Several ways to predict future ratings**
  - Simple approach: popularity sorting, High Avg. rating, …
    - Simple and easy-to-implement
    - Global recommendation (non-personalized)

	5		8	7		
	9	10	8	9		
	6	6		6	10	7
	8				6	10

Problem Setting

- **This is a typical recommendation scenario**
- **Several ways to predict future ratings**
  - Heuristic approach: Neighborhood-based
    - Looking for the **most similar K users**
    - Aggregating their ratings on a target movie

$$(a)\ r_{u,i} = \frac{1}{|N|}\sum_{u' \in N} r_{u',i}$$

$$(b)\ r_{u,i} = \frac{1}{k}\sum_{u' \in N} sim(u,u') \times r_{u',i}$$

$$(c)\ r_{u,i} = \bar{r}_u + \frac{1}{k}\sum_{u' \in N} sim(u,u') \times (r_{u',i} - \bar{r}_{u'})$$

5		8	7		
9	10	8	9		
6	6		6	10	7
8				6	10

- N: Set of neighbors for user u
- $sim(u,u')$: similarity between users
- $\bar{r}_u$: Average of ratings of user u

Problem Setting

- **This is a typical recommendation scenario**
- **Several ways to predict future ratings**
  - Machine learning approach?
    - What is ground truth?: observed ratings
    - What is our model?: **Matrix factorization**! (a.k.a. latent factor models)

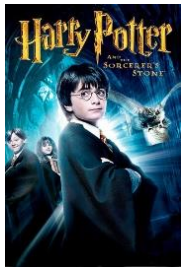
	5		8	7		
	9	10	8	9		
	6	6		6	10	7
	8				6	10

Matrix Factorization

- **Ratings come from interaction between latent factors**
  - Assumption: there are **user latent factors** and **item latent factors** that result in the rating score.

[a, b, c]
<user latent factors>

5



[a′, b′, c′]

3

[a′, b′, c′]

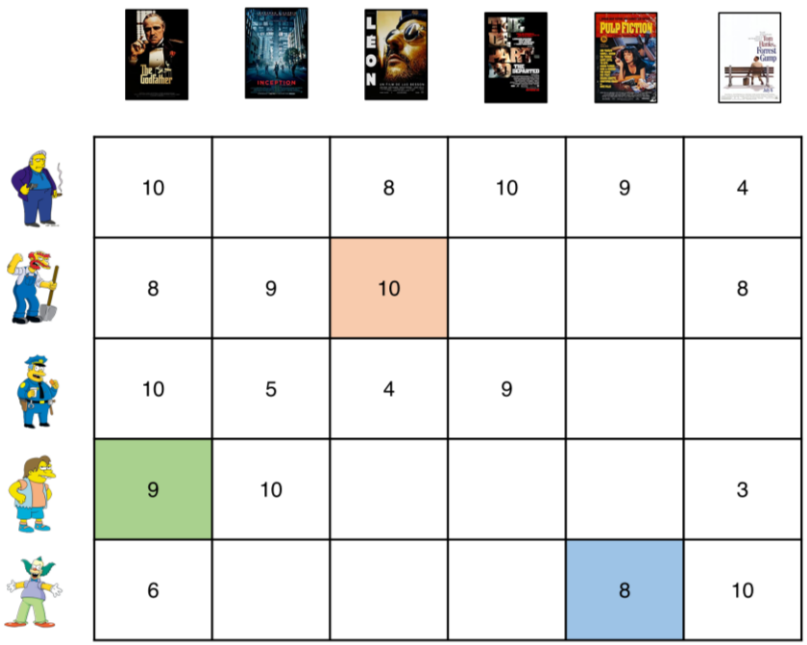
[a, b, c] x [a′, b′, c′] = 5

[a, b, c] x [a′, b′, c′] = 3

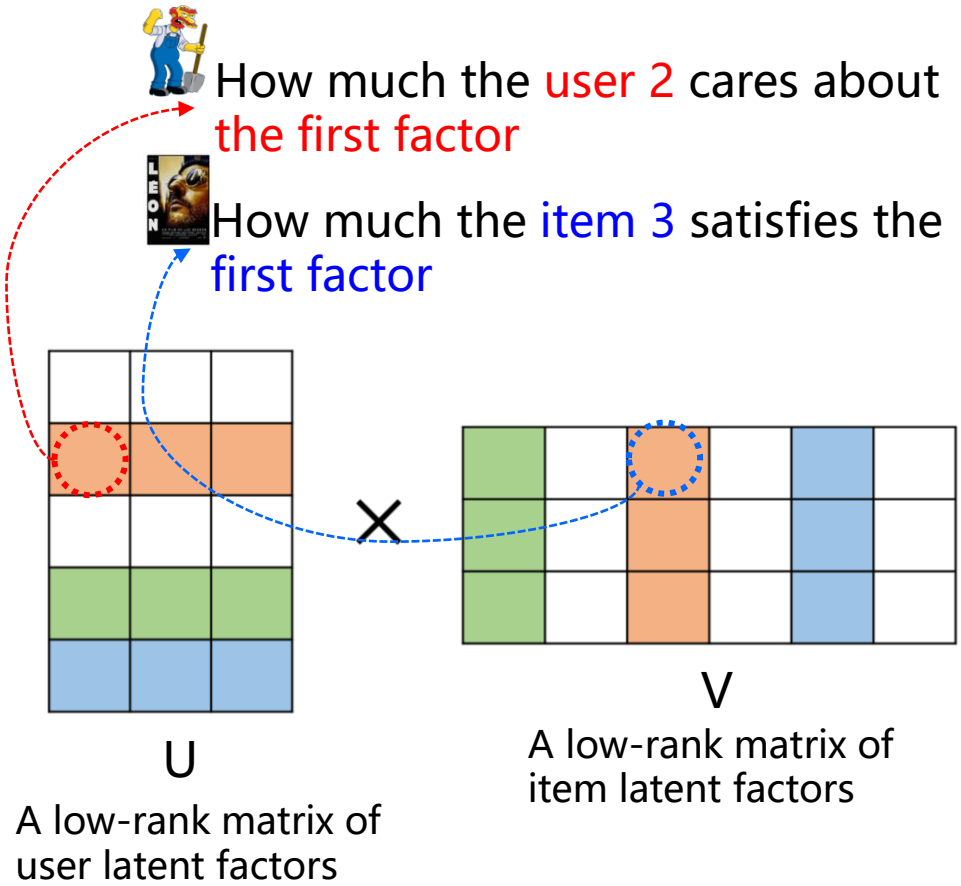
<item latent factors> <linear interaction> <ratings>

Matrix Factorization

- **Latent factors**
  - Features that describe characteristics of users and items **hidden** in data
  - We don't know the explicit meaning of each factor in the latent matrices, but there might be "some" meaning

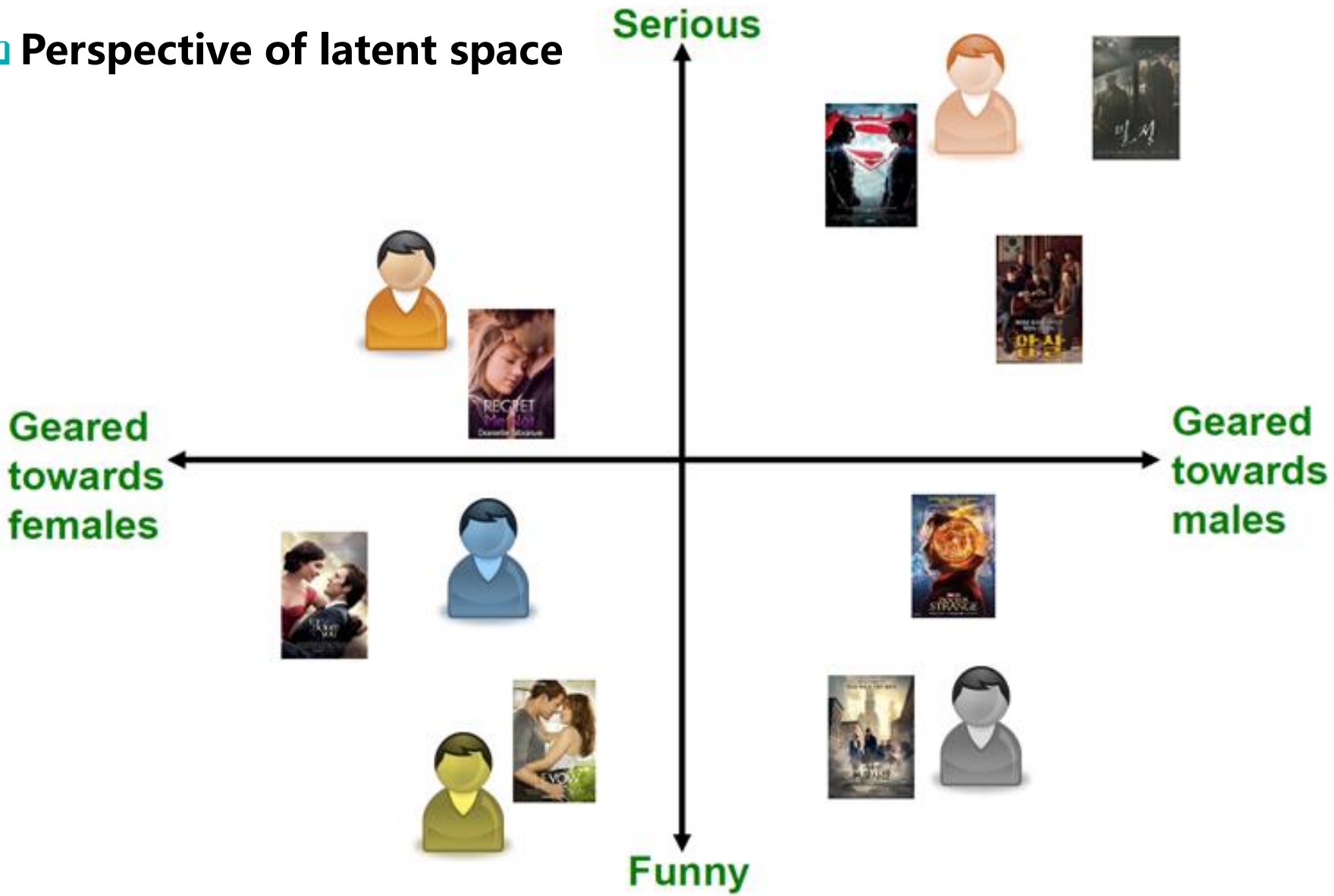


≈



Matrix Factorization

- **Perspective of latent space**



Matrix Factorization

- **How can I know such latent factor values?**
  - "**Learn**" the latent factors, U and V, that may originate the ratings
  - All the latent factors in U and V are **model parameters**
  - **Randomly initialized**, and **learned** through some **optimization** techniques.

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3			5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

R

Objective:

$$\min_{U,V} \sum_{u=1}^{m} \sum_{i=1}^{n} y_{ui}\left(r_{uj} - U_u V_i^T\right)^2$$

$$y_{ui} \begin{cases} 1 & if\ r_{ui}\ \text{exists} \\ 0 & \text{otherwise} \end{cases}$$

≈

factors

users

0.1	-0.4	0.2
-0.5	0.6	0.5
-0.2	0.3	0.5
1.1	2.1	0.3
-0.7	2.1	-0.2
-1.0	0.7	0.3

U

items

1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

factors

V^T

Matrix Factorization

- **Rating prediction with the completely learned U and V**
  - We can reconstruct "**dense**" rating matrix through $U \cdot V^T$

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3		??	5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

$$\tilde{r}_{ui} = U_u V_i^T = \sum_{f=1}^{k} U_{uf} V_{fi}^T$$

U_u = row u of U
V_i^T = column i of V

R ≈ **U** · **V**^{T}

factors

0.1	-0.4	0.2
-0.5	0.6	0.5
-0.2	0.3	0.5
1.1	2.1	0.3
-0.7	2.1	-0.2
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users

items

1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

factors

Matrix Factorization

- **Objective function: minimizing the sum of squared error**

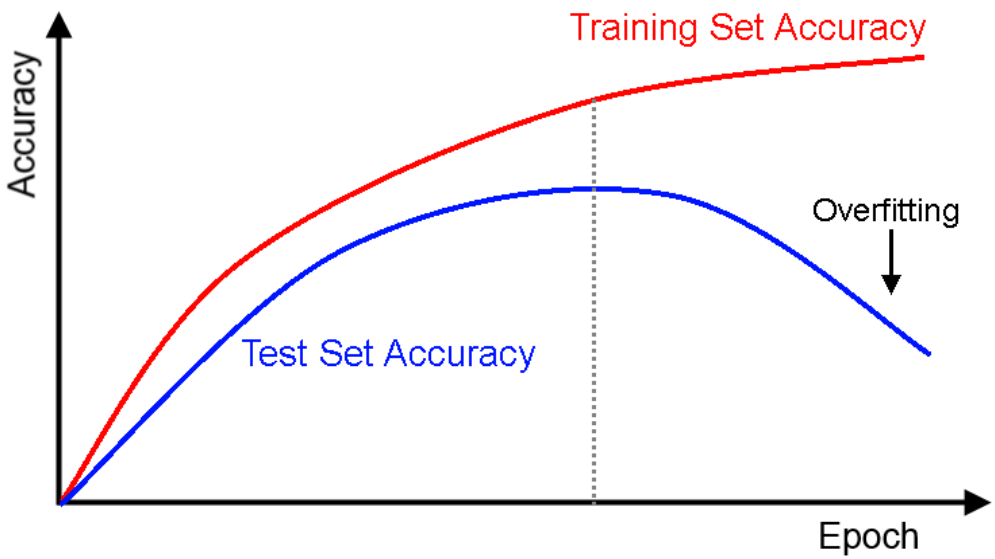
$$\underset{U,V}{\operatorname{argmin}} \sum_{u=1}^{m} \sum_{i=1}^{n} y_{ui} \left(r_{ui} - U_u V_i^T \right)^2$$

- **Practically: there is an overfitting issue**

0.1	-0.4	0.2
-0.5	0.6	0.5
-0.2	100	0.5
1.1	2.1	0.3
-0.7	2.1	-0.2
-1.0	0.7	0.3

U

1.1	-0.2	0.3	0.5	-2.0
-0.8	0.7	0.5	1.4	-81
2.1	-0.4	0.6	1.7	2.4



Matrix Factorization

- **Objective function with the regularization term**

Goodness of fit Regularization

$$\underset{U,V}{\operatorname{argmin}} \sum_{u=1}^{m} \sum_{i=1}^{n} y_{ui}\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\left|\left|U\right|\right|^2 + \left|\left|V\right|\right|^2\right)$$

  - The goodness of fit is to reduce the prediction error
  - The regularization term is used to alleviate the overfitting problem

- **Two ways to training the MF model**
  - Stochastic gradient descent (**SGD**)
    - Training U and V simultaneously
  - Alternating least squares (**ALS**)
    - Fix one of U and V, and then optimize the other

Stochastic Gradient Descent

Objective:

Goodness of fit — Regularization

$$\underset{U,V}{\operatorname{argmin}} \sum_{u=1}^{m}\sum_{i=1}^{n} y_{ui}\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\|U\|^2 + \|V\|^2\right)$$

Process:

1. Initialize U and V randomly
2. Repeat:
   1. Choose a rating **(r, u, i)** randomly
   2. Then the loss on the chosen (u, i) is:

$$\frac{1}{2}\left(\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\|U\|^2 + \|V\|^2\right)\right)$$

   3. Update via
      - $e_{ui} = r_{ui} - U_u V_i^T$
      - $U_u \leftarrow U_u + \eta\left(e_{ui} V_i^T - \lambda U_u\right)$
      - $V_i \leftarrow V_i + \eta\left(e_{ui} U_u - \lambda V_i\right)$
      - Update U_u and V_i iteratively.

They are the partial derivative of U_u and V_i.

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3			5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

R

$$\min_{U,V} \sum_{u=1}^{m}\sum_{i=1}^{n} y_{ui}\left(r_{uj} - U_u V_i^T\right)^2$$

$$y_{ui} \begin{cases} 1 & \text{if } r_{ui} \text{ exists} \\ 0 & \text{otherwise} \end{cases}$$

≈

users — factors

0.1	-0.4	0.2
-0.5	0.6	0.5
-0.2	0.3	0.5
1.1	2.1	0.3
-0.7	2.1	-0.2
-1.0	0.7	0.3

U

items — factors

1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

V^T

Stochastic Gradient Descent

$$e_{ui} = r_{ui} - U_u V_i^T$$

- **Partial derivative of the loss** $L = \frac{1}{2}\left(\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\|U\|^2 + \|V\|^2\right)\right)$

$$\frac{\partial L}{\partial U_u} = \frac{1}{2}\left(2e_{ui} \cdot e'_{ui} + 2\lambda U_u\right)$$

$$= e_{ui} \cdot e'_{ui} + \lambda U_u$$

$$= e_{ui} \cdot (-V_i) + \lambda U_u$$

$$\frac{\partial L}{\partial V_i} = \frac{1}{2}\left(2e_{ui} \cdot e'_{ui} + 2\lambda V_i\right)$$

$$= e_{ui} \cdot e'_{ui} + \lambda V_i$$

$$= e_{ui} \cdot (-U_u) + \lambda V_i$$

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3			5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

R

$$\min_{U,V} \sum_{u=1}^{m} \sum_{i=1}^{n} y_{ui}\left(r_{uj} - U_u V_i^T\right)^2$$

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factors

≈ users

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1.1	2.1	0.3
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U

items

1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

factors

V^T

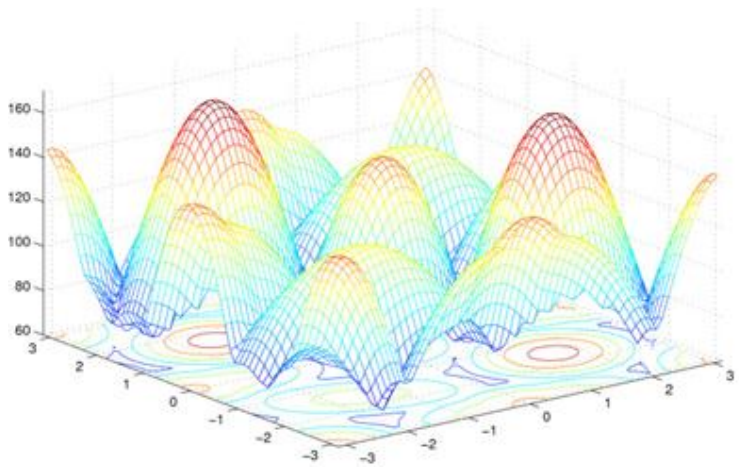
- $U_u \leftarrow U_u + \eta\left(e_{ui} V_i^T - \lambda U_u\right)$
- $V_i \leftarrow V_i + \eta\left(e_{ui} U_u - \lambda V_i\right)$
- Update U_u and V_i iteratively.

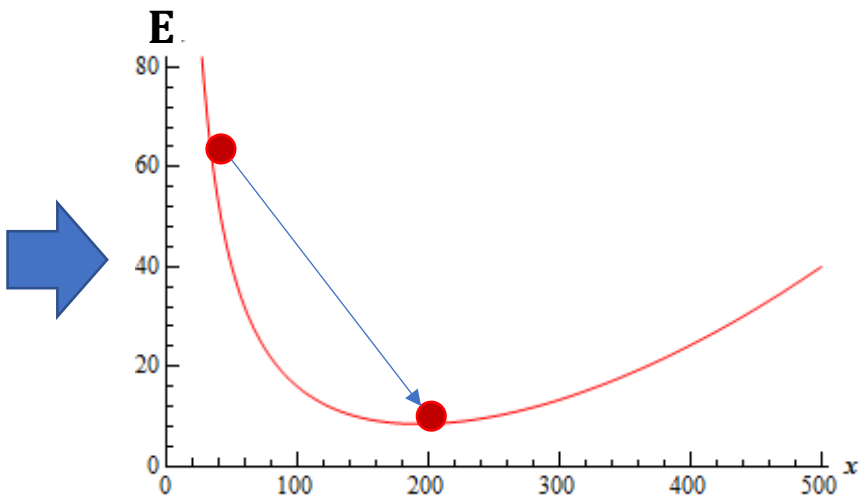
Alternating Least Squares

- **Fix one, then the loss surface becomes simple**

Goodness of fit **Regularization**

$$\underset{U,V}{\operatorname{argmin}} \sum_{u=1}^{m}\sum_{i=1}^{n} y_{ui}\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\left|\left|U\right|\right|^2 + \left|\left|V\right|\right|^2\right)$$





$$\frac{\partial E}{\partial U_i} = 0$$

Alternating Least Squares

- **Objective:**

Goodness of fit Regularization

$$\underset{U,V}{\operatorname{argmin}} \sum_{u=1}^{m} \sum_{i=1}^{n} y_{ui}\left(r_{ui} - U_u V_i^T\right)^2 + \lambda\left(\left|\left|U\right|\right|^2 + \left|\left|V\right|\right|^2\right)$$

- **Procedure**

1. Initialize the user latent factors U randomly
2. For each item i, let r_i be the vector of ratings of that item. Compute:

$$V_i^T = (U^T U + \lambda I)^{-1} U^T r_i \quad \text{(here, } U \text{ is constant)}$$

3. For each user u, let r_u be the vector of ratings of that item. Compute:

$$U_u^T = (V^T V + \lambda I)^{-1} V^T r_u \quad \text{(here, } V \text{ is constant)}$$

4. Repeat 2), 3) until convergence

Alternating Least Squares

- **Partial derivative of the loss**
  - When U is constant, $L = \left(r_i - UV_i^T\right)^2 + \lambda\left(\left|\left|U\right|\right|^2 + \left|\left|V\right|\right|^2\right)$

$$L = \left(r_i - UV_i^T\right)^T\left(r_i - UV_i^T\right) + \lambda V_i V_i^T$$

Let $V_i^T = \beta$. Then,

$$L = (r_i - U\beta)^T(r_i - U\beta) + \lambda\beta^T\beta$$

Let $\frac{\partial L}{\partial \beta} = 0$. Then,

$$\beta = V_i^T = (U^TU + \lambda I)^{-1}U^Tr_i$$

repeat i=1 to n

$$*Q(\beta) = (y - X\beta)^T\ (y - X\beta) + \lambda\beta^T\beta$$

$$= y^Ty\ - 2\beta^TX^Ty + \beta^T\left(X^TX + \lambda I_p\right)\beta$$

$$\frac{\partial}{\partial\beta}Q(\beta) = 2X^Ty + 2\left(X^TX + \lambda I_p\right)\beta = 0$$

$$\hat{\beta}^{ridge} = \left(X^TX + \lambda I_p\right)^{-1}X^Ty$$

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3			5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

R

$$\min_{U,V}\sum_{u=1}^{m}\sum_{i=1}^{n} y_{ui}\left(r_{uj} - U_uV_i^T\right)^2$$

$$y_{ui}\begin{cases}1 & if\ r_{ui}\ \text{exists}\\ 0 & \text{otherwise}\end{cases}$$

factors

≈ users

0.1	-0.4	0.2
-0.5	0.6	0.5
-0.2	0.3	0.5
1.1	2.1	0.3
-0.7	2.1	-0.2
-1.0	0.7	0.3

U

items

factors

1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

V^T

Alternating Least Squares

- **Partial derivative of the loss**
  - When V is constant, $L = \left(r_u - U_u V^T\right)^2 + \lambda\left(\|U\|^2 + \|V\|^2\right)$

Let $r_u = R_u^T$. Then, we can re-write the loss:

$$L = (r_u - VU_u^T)^T(r_u - VU_u^T) + \lambda U_u U_u^T$$

Let $U_u^T = \beta$. Then,

$$L = (r_u - V\beta)^T(r_u - V\beta) + \lambda\beta^T\beta$$

Let $\frac{\partial L}{\partial \beta} = 0$. Then,

$$\beta = U_u^T = (V^TV + \lambda I)^{-1}V^T r_u$$

repeat u=1 to m

$$*Q(\beta) = (y - X\beta)^T\ (y - X\beta) + \lambda\beta^T\beta$$

$$= y^Ty\ - 2\beta^TX^Ty + \beta^T\left(X^TX + \lambda I_p\right)\beta$$

$$\frac{\partial}{\partial\beta}Q(\beta) = 2X^Ty + 2\left(X^TX + \lambda I_p\right)\beta = 0$$

$$\hat{\beta}^{ridge} = \left(X^TX + \lambda I_p\right)^{-1}X^Ty$$

	i_1	i_2	i_3	i_4	i_5	i_6	i_7	i_8	i_9	i_{10}	i_{11}	i_{12}
u_1	1		3			5			5		4	
u_2			5	4			4			2	1	3
u_3	2	4		1	2		3		4	3	5	
u_4		2	4		5			4			2	
u_5			4	3	4	2					2	5
u_6	1		3		3			2			4	

R

$$\min_{U,V} \sum_{u=1}^{m}\sum_{i=1}^{n} y_{ui}\left(r_{uj} - U_u V_i^T\right)^2$$

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factors

≈ users

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U

items

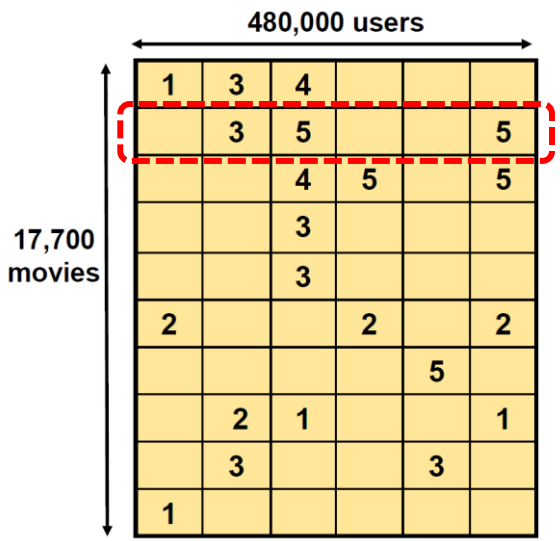
1.1	-0.2	0.3	0.5	-2.0	-0.5	0.8	-0.4	0.3	1.4	2.4	-0.9
-0.8	0.7	0.5	1.4	0.3	-1.0	1.4	2.9	-0.7	1.2	-0.1	1.3
2.1	-0.4	0.6	1.7	2.4	0.9	-0.3	0.4	0.8	0.7	-0.6	0.1

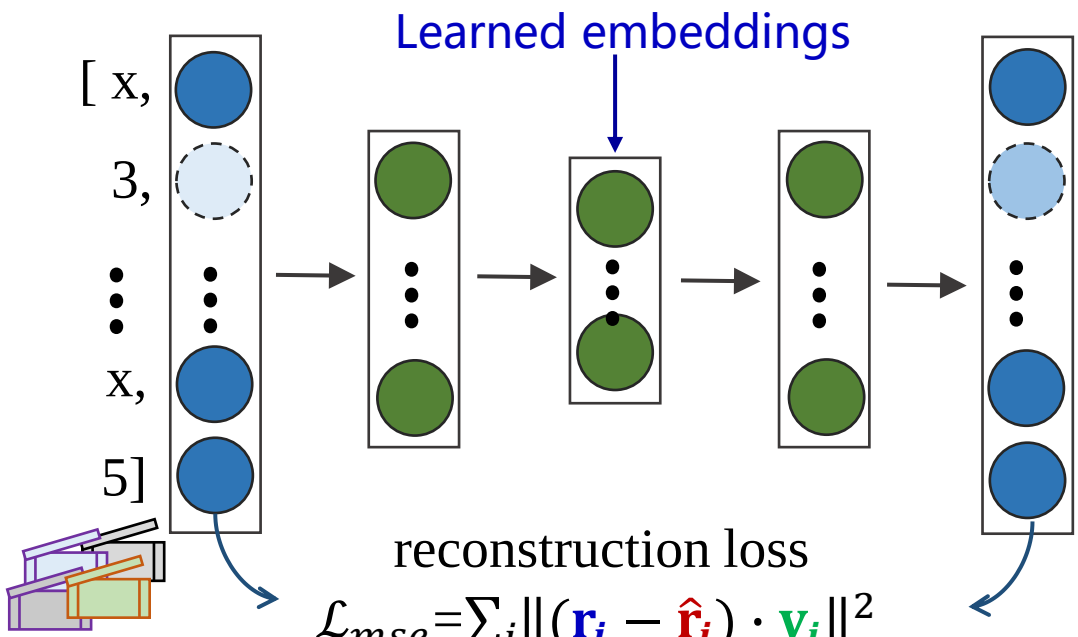
factors

V^T

Autoencoder for MF

- **Same objective function, different model**
  - Matrix factorization => Autoencoder
- **Two types of Autoencoders for recommendation**
  - user-based and item-based
- **Users and items are represented as vectors**





reconstruction loss

$$\mathcal{L}_{mse}=\sum_i \|(\mathbf{r}_i - \hat{\mathbf{r}}_i)\cdot \mathbf{y}_i\|^2$$

Actual ratings — Predicted ratings — Indicator {0, 1}

Thank You